

Universal Properties from Quantum Many-Body Dynamics

Master's Thesis Defence

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Barcelona Supercomputing Center · September 2026



**Barcelona
Supercomputing
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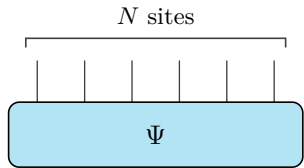
Centro Nacional de Supercomputación



**Master in Quantum
Science and Technology
Barcelona**

Simulating Quantum Many-Body Systems

motivation



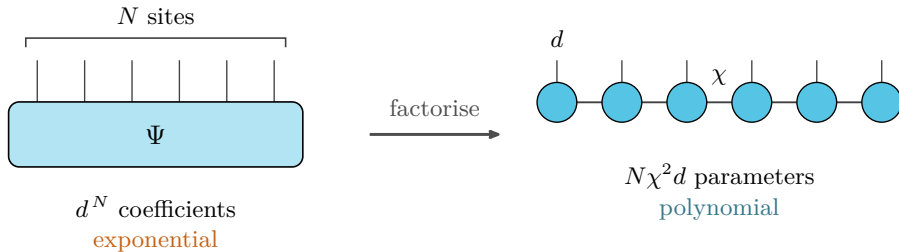
d^N coefficients
exponential

$$|\Psi\rangle = \sum_{i_1 \dots i_N} \Psi_{i_1 i_2 \dots i_N} |i_1 i_2 \dots i_N\rangle$$

the **many-body problem**: a state is a list of d^N numbers

Simulating Quantum Many-Body Systems

motivation

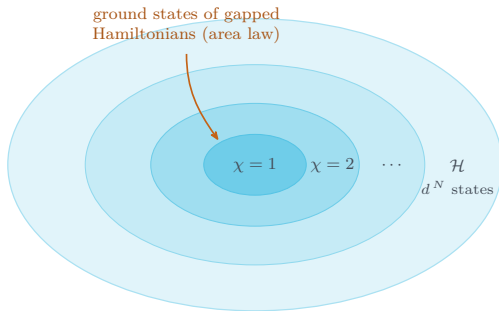


$$\psi_{i_1 i_2 \dots i_N} = \sum_{\alpha_1 \dots \alpha_{N-1}} A_{\alpha_1}^{i_1} A_{\alpha_1 \alpha_2}^{i_2} \dots A_{\alpha_{N-1}}^{i_N}$$

a **matrix product state (MPS)**: χ = the entanglement it can carry = the memory it costs

Simulating Quantum Many-Body Systems

motivation



$\chi = d^{N/2}$ already reaches every state —
 χ counts Schmidt values across *one* cut

a small χ already describes states with low entanglement
and fast-decaying correlations — the **area law**

Reading a Tensor Network

motivation



$$\sum_{\alpha} A_{i\alpha} B_{\alpha j} = C_{ij}$$

joining two legs = summing over that index = a **contraction**

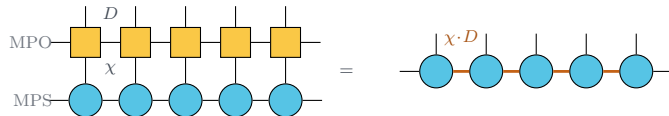
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an **MPO**: an operator in the same language

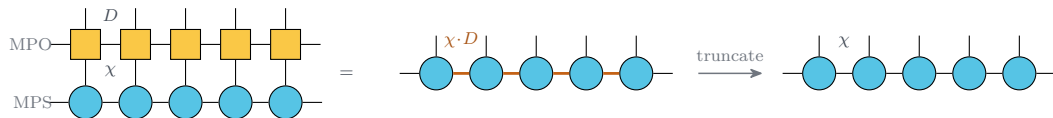
Reading a Tensor Network

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$$\sum_{\alpha} A_{i\alpha} B_{\alpha j} = C_{ij}$$

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an **MPO**: an operator in the same language

keep the largest Schmidt values

Dynamics: the Entanglement Barrier

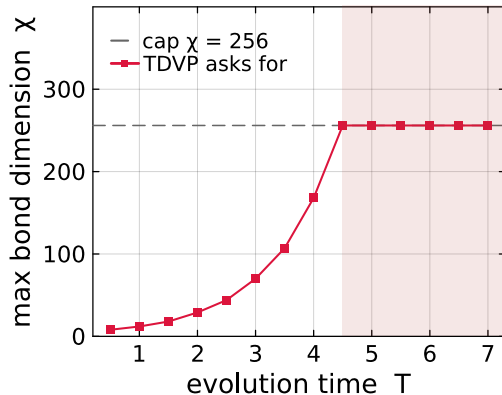
motivation

dynamics runs into a different problem:

$$\chi \xrightarrow{\text{apply MPO}} D\chi \xrightarrow{\text{truncate}} \chi$$

after a quench:

$$S_x(t) \sim t \implies \chi \sim e^{ct}$$



Dynamics: the Entanglement Barrier

motivation

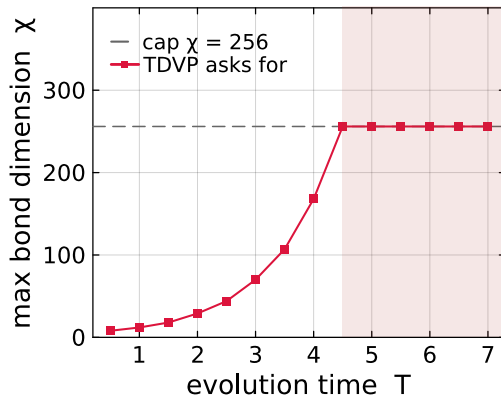
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after a quench:

$$S_x(t) \sim t \implies \chi \sim e^{ct}$$

**this work: time evolution
that avoids the barrier,
for critical quenches in 1D**



What We Are Trying to Extract

motivation

at a critical point the microscopic details stop mattering, and the model is described by a set of **universal numbers**

$$c = \frac{1}{2}$$

the **central charge**

$$x_\alpha \in \left\{ \frac{1}{2}, \frac{3}{2}, 2, \frac{5}{2}, \dots \right\}$$

the **boundary operators'** dimensions

the numbers the **conformal field theory (CFT)** of the Ising class predicts

read them off a real-time dynamical object at the critical point

already done for **exactly solvable** models — we target the **non-integrable** ones

Carignano & Tagliacozzo, *Quantum* (2025)

transverse contraction reaches long evolution times

- time-evolve a **non-integrable** chain and compute the Loschmidt echo
 - extract its **universal data** from the simulations
 - compare with the **CFT predictions**

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- compare with the **CFT predictions**

1. the Loschmidt echo and the CFT mapping
2. transverse contraction
3. the model and its equilibrium properties ★
4. a translation-invariant evolution operator ★
5. power iterations: the dominant pair
6. the block power method: the tower ★
7. results

The Loschmidt Echo

1 · the echo

$$\mathcal{L}(\tau) = \langle \Psi_0 | e^{-\tau H} | \Psi_0 \rangle \quad \tau = \beta_0 + iT$$

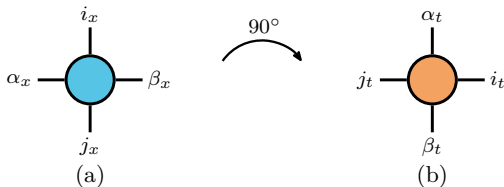
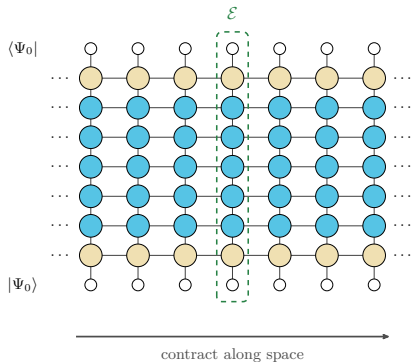
$$e^{-iHT} \xrightarrow{\text{rotation } T \rightarrow -i\tau} e^{-\tau H} \quad \text{Euclidean partition function — where the CFT applies}$$

strip width $\beta = v(2\beta_0 + iT)$, growing with time

Carignano & Tagliacozzo, *Quantum* (2025)

The Transverse Rotation

2 · transverse contraction



virtual space $\rightarrow$ physical *time*
physical space $\rightarrow$ virtual direction

translation invariance $\Rightarrow$ every column is the same operator: the transfer matrix $\mathcal{E}$

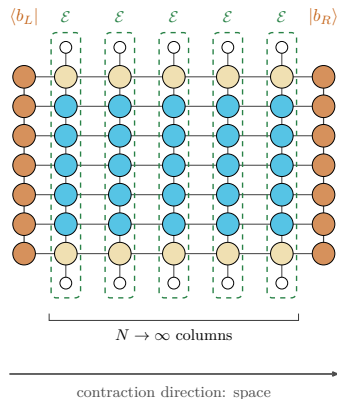
Bañuls et al. (2009) · code: `ITransverse.jl`, Carignano (2026)

Dynamics as an Eigenproblem

2 · transverse contraction

$$\begin{aligned}\mathcal{L}(T) &= \langle b_L | \mathcal{E}^N | b_R \rangle = \sum c_i \mu_i^N \\ &= c_0 \mu_0^N \underbrace{\left[1 + \sum_{i \geq 1} \frac{c_i}{c_0} \left(\frac{\mu_i}{\mu_0} \right)^N \right]}_{\rightarrow 1}\end{aligned}$$

$$\begin{aligned}\mathcal{E} |r_i\rangle &= \mu_i |r_i\rangle, & \langle \ell_i | \mathcal{E} &= \mu_i \langle \ell_i | \\ c_i &= \langle b_L | r_i \rangle \langle \ell_i | b_R \rangle\end{aligned}$$



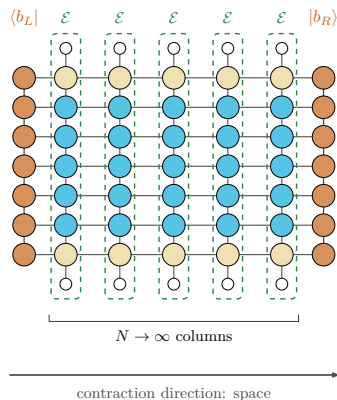
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$\mathcal{E}\mathcal{E}^\dagger \neq \mathcal{E}^\dagger\mathcal{E}$: everything reduces to the dominant pair and to the spectrum of the non-Hermitian $\mathcal{E}$



b_L, b_R : random seeds — repeated application *projects* them onto the dominant pair,
 $\langle b_L | \mathcal{E}^N \rightarrow \langle L |, \mathcal{E}^N | b_R \rangle \rightarrow | R \rangle$

at a critical point the continuum theory is a
conformal field theory (CFT)

↓ so many symmetries that:

the whole model is fixed by
one number: the central charge c

each operator is fixed by
one number: its dimension x

What a CFT Gives Us

2 · transverse contraction

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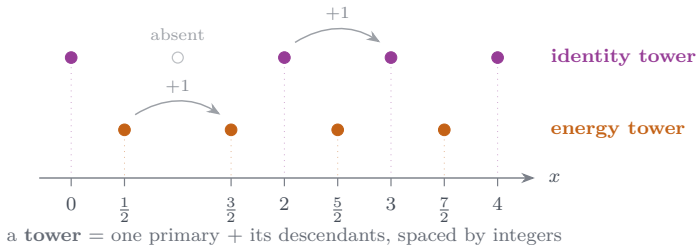
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map the plane to a **strip**
with edges and some of the
operators go missing

each dimension fixes how fast
that operator's correlations die at
the edge



after the Wick rotation: a CFT on a strip of width $\beta = \nu(2\beta_0 + iT)$

$$\mathcal{E} = e^{-\hat{H}} \quad (a=1), \quad \hat{H} = \frac{\pi}{\beta} \left(L_0 - \frac{c}{24} \right) \quad \implies \quad \mu_i = \exp \left[-\frac{\pi}{\beta} \left(x_i - \frac{c}{24} \right) \right]$$

$\mathcal{E}$ steps one spacing *along* the strip $\Rightarrow$ a weight, not a phase
 L_0 : generator of scalings, eigenvalues x_i · $-c/24$: the Casimir shift

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1. emergent dual unitarity — the moduli lock:

$$\frac{|\mu_i|}{|\mu_0|} = 1 - \frac{b_i}{T^2} \longrightarrow 1$$

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1. emergent dual unitarity — the moduli lock:

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2. the dominant phase — the expression we fit:

$$\frac{\text{Im } \lambda_0}{T} = a_0 + \frac{B}{T} + \frac{C}{T^2}, \quad B = -\pi \text{ (branch, fixed)} \quad \Longrightarrow \quad c = \frac{24 \nu |C|}{\pi}$$

$\lambda_0 = \log(-\mu_0)$, and $\text{Im } \log \mu = \arg \mu$ — the phase

CFT Predictions II — the Operator Content

2 · transverse contraction

$$\Delta\theta_\alpha = \arg \mu_\alpha - \arg \mu_0 = \frac{\pi \textcolor{brown}{V} x_\alpha}{T}$$

each subleading eigenvalue

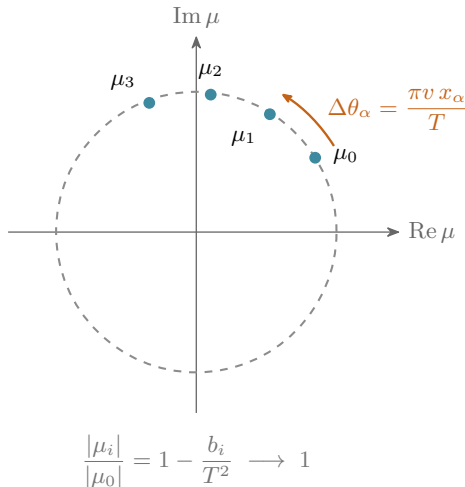
↔ one boundary operator

so for the Ising class we should see

μ_0 , then one eigenvalue per

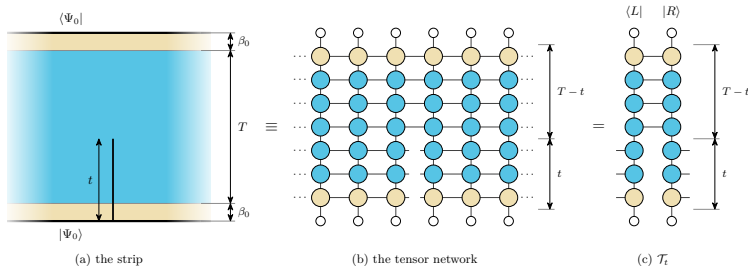
$$x_\alpha \in \left\{ \frac{1}{2}, \frac{3}{2}, 2, \frac{5}{2}, \dots \right\}$$

Cardy, *Nucl. Phys. B* (1986)

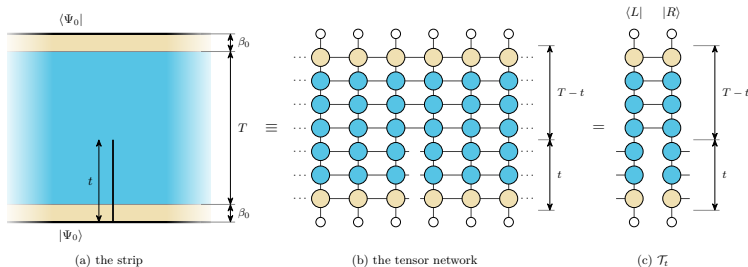


cost of the contraction = entanglement across a cut in **time**

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$$\mathcal{T}_t = \frac{\text{Tr}_{>t} |R\rangle \langle L|}{\langle L|R\rangle}$$

$$S_2^{\text{gen}}(t) = -\ln \text{Tr}(\mathcal{T}_t^2)$$

$$\mathrm{Re} S_2^{\mathrm{gen}}(t) = s_0 + \frac{c}{8} \log \left[\frac{2T}{\pi} \sin \frac{\pi t}{T} \right] \quad (\text{chord dome})$$

$$\mathrm{Im} S_2^{\mathrm{gen}}(t) = \frac{\pi c}{16} \quad (\text{constant plateau})$$

Calabrese & Cardy, *J. Stat. Mech.* (2004)

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logarithmic in $T \Rightarrow$ polynomial resources

Calabrese & Cardy, *J. Stat. Mech.* (2004)

The ANNNI-type Chain Stays in the Ising Class

3 · the model

$$H = - \sum_i \left[\underbrace{\sigma_i^z \sigma_{i+1}^z + \lambda \sigma_i^x}_{\text{transverse-field Ising}} + p \underbrace{(\sigma_i^z \sigma_{i+2}^z + \lambda \sigma_i^x \sigma_{i+1}^x)}_{\text{new with } p} \right], \quad \lambda = 1$$

after a Jordan–Wigner transformation:

$p = 0$: **free** fermions — solvable exactly

$p > 0$: **interacting** fermions

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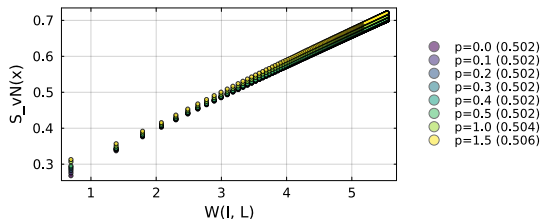
but is it still Ising?

ask the ground state:

$$S_{\text{vN}}(l, L) = \frac{c}{6} \underbrace{\log \left[\frac{2L}{\pi} \sin \frac{\pi l}{L} \right]}_W + k$$

DMRG, $N = 400$

$c = 0.50$ **for every** p



The Sound Velocity

3 · the model

diagonalise small rings of N sites
exactly:

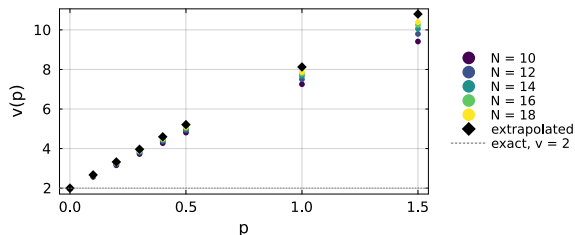
$$v(N) = \frac{N}{2\pi} [E_1(2\pi/N) - E_1(0)]$$

$$v(N) = v_\infty + \frac{C}{N^2}$$

finite-size correction, exponent fixed
empirically ($v = 2$ at $p = 0$ as
reference)

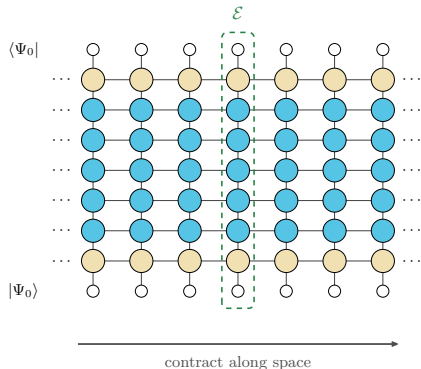
estimator: Alcaraz, *Phys. Rev. B* (2016)

$v T =$ conformal distance in the strip



The Time Evolution Operator

4 · methods



translation invariance $\Rightarrow$
 $U(\delta t)$ built from one repeated bulk
tensor W

second order in δt

$$H = - \sum_i \left[\sigma_i^z \sigma_{i+1}^z + \lambda \sigma_i^x + p(\sigma_i^z \sigma_{i+2}^z + \lambda \sigma_i^x \sigma_{i+1}^x) \right]$$

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$$W_H = \begin{pmatrix} \mathbb{1} & C & D \\ 0 & A & B \\ 0 & 0 & \mathbb{1} \end{pmatrix}$$

$$= \begin{pmatrix} \mathbb{1} & -\sigma^z & -p\lambda \sigma^x & -p\sigma^z & -\lambda \sigma^x \\ 0 & 0 & 0 & 0 & \sigma^z \\ 0 & 0 & 0 & 0 & \sigma^x \\ 0 & \mathbb{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbb{1} \end{pmatrix}$$

$$D_W = 1 + 3 + 1 = 5$$

idle · three memory channels · done

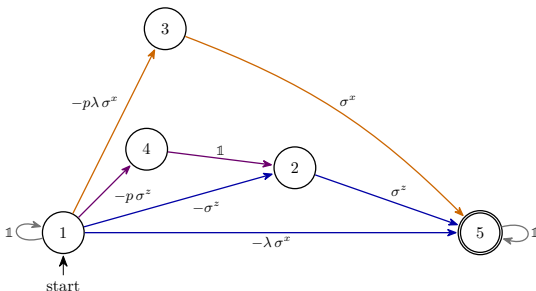
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idle · three memory channels · done



directional machine $\Rightarrow \mathcal{E}$ **non-normal**

Exponentiating the MPO: the VD2 Construction

4 · methods

$$e^{\tau H} = \mathbb{1} + \tau \sum_x H_x + \frac{\tau^2}{2} \underbrace{\sum_{x,y} H_x H_y}_{\text{including the overlapping pairs}} + \dots$$

$$\tau = -i \delta t$$

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keeping the overlapping pairs is what makes it second order

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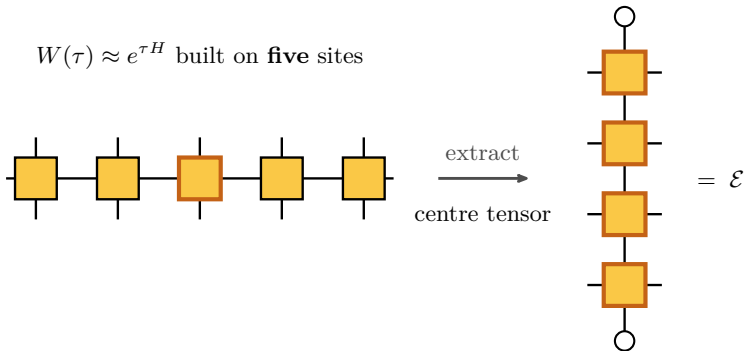
keeping the overlapping pairs is what makes it second order

a modification of the finite-state machine: **two machines with two indices**,
so two terms can be in flight at once

$$\text{bond} = 1 + \chi_H + \chi_H^2 = 13$$

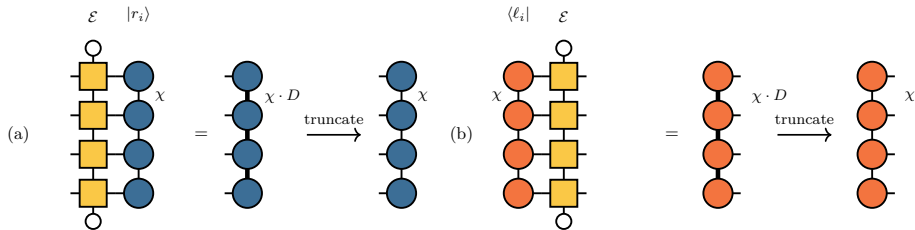
Van Damme et al., *SciPost Phys.* **17**, 135 (2024)

One Tensor from the Bulk



The Power Method: the Dominant Pair

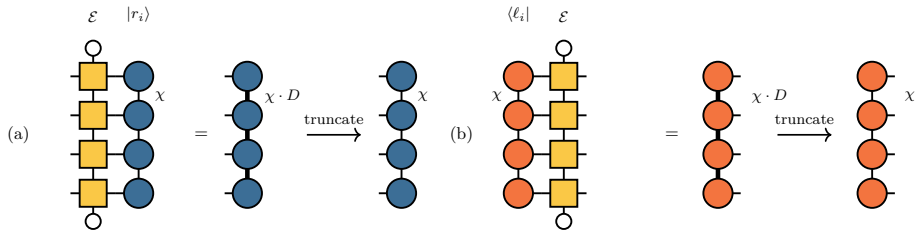
$$\mathcal{E} |r_i\rangle = \mu_i |r_i\rangle, \quad \langle \ell_i | \mathcal{E} = \mu_i \langle \ell_i |, \quad \langle \ell_i | r_j \rangle = \delta_{ij}$$



The Power Method: the Dominant Pair

4 · methods

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output:

1. the dominant pair $\langle L |, |R \rangle$

$$2. \mu_0 = \frac{\langle L | \mathcal{E} | R \rangle}{\langle L | R \rangle}$$

validity: random complex seeds · independent seeds agree · phase continuous in T

the same iteration, on a **block** of k states per side

$$\sum_j M_{ij} v_j = \theta \sum_j S_{ij} v_j, \quad S_{ij} = \langle L_i | R_j \rangle, \quad M_{ij} = \langle L_i | \mathcal{E} | R_j \rangle$$

a $k \times k$ generalised eigenproblem, projected onto the block

the same iteration, on a **block** of k states per side

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a $k \times k$ generalised eigenproblem, projected onto the block

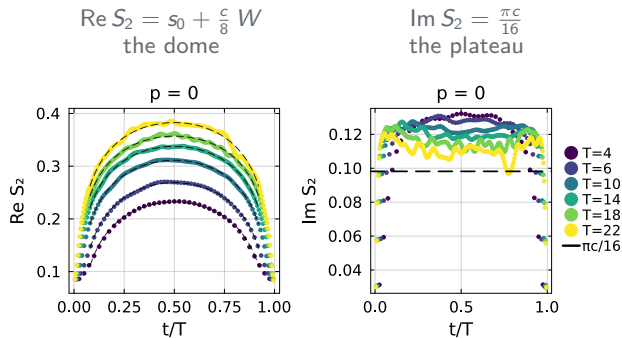
$\theta_0, \dots, \theta_{k-1} \approx \mu_0, \dots, \mu_{k-1}$: the k leading eigenvalues,
and from their phases the gaps $\Delta\theta_\alpha$

the four predictions, measured at every coupling

p	v_∞	from the eigenvectors		from the eigenvalues		
		window	c (entropy)	window	c (spectral)	x_1
0	2.00	$4 \leq T \leq 22$	0.49 ± 0.01	$2 \leq T \leq 22$	0.45	0.498
0.1	2.67	$4 \leq T \leq 13$	0.50 ± 0.04	$2 \leq T \leq 17$	0.47	0.497
0.3	3.97	$2 \leq T \leq 9$	0.53 ± 0.04	$2 \leq T \leq 10$	0.49	0.493
0.5	5.21	$2 \leq T \leq 5$	0.58 ± 0.04	$2 \leq T \leq 6$	0.53	0.487

Results I — the Entropy Route at $p = 0$

5 · results



finite- T correction and extrapolation:

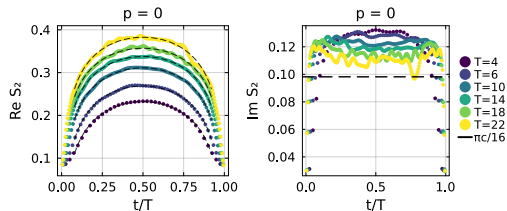
$$\text{plateau}(T) = a + b T^{-1/2} \Rightarrow c = \frac{16 a}{\pi}$$

the error bar is the seed-to-seed spread

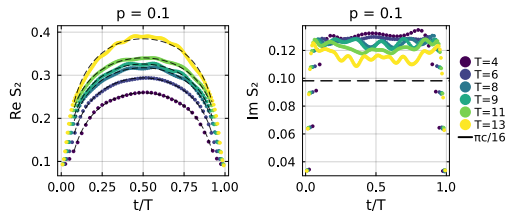
p	window	c (entropy)
0	$4 \leq T \leq 22$	0.49 ± 0.01

Results I — the Entropy Route at Every Coupling

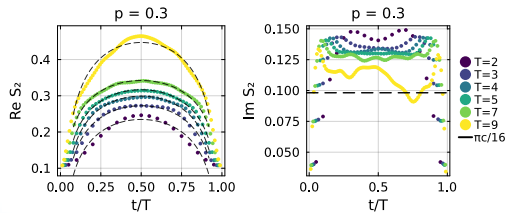
5 · results



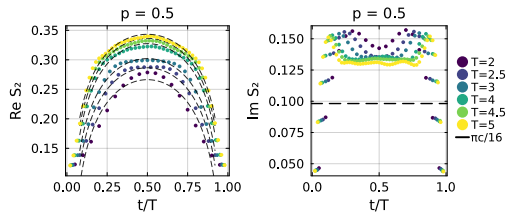
$c = 0.49 \pm 0.01$ ($4 \leq T \leq 22$, $v_\infty T = 8-44$)



$c = 0.50 \pm 0.04$ ($4 \leq T \leq 13$, $v_\infty T = 11-35$)



$c = 0.53 \pm 0.04$ ($2 \leq T \leq 9$, $v_\infty T = 8-36$)



$c = 0.58 \pm 0.04$ ($2 \leq T \leq 5$, $v_\infty T = 10-26$)

$|\mu_0|$ **constant**
while the phase winds

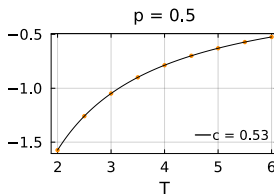
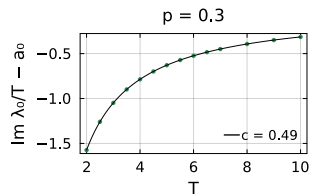
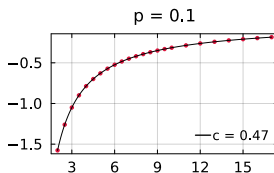
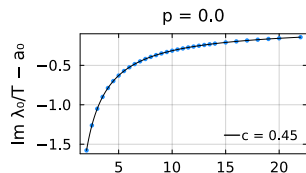
$$\frac{|\mu_i|}{|\mu_0|} = 1 - \frac{b_i}{T^2} \rightarrow 1$$

Carignano & Tagliacozzo (2025)

animated in Acrobat; static frame elsewhere

Results III — the Central Charge from the Phase

5 · results



$$\frac{\text{Im } \lambda_0}{T} = a_0 + \frac{B}{T} + \frac{C}{T^2}$$

a_0, C free

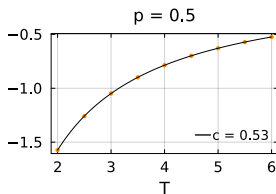
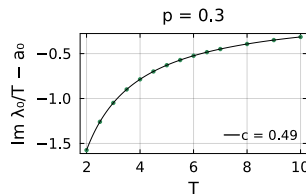
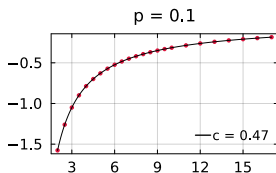
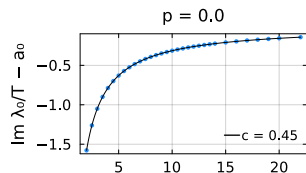
$B = -\pi$ fixed (branch)

$$c = 24 \nu |C|/\pi$$

p	window	c (spectral)
0	$2 \leq T \leq 22$	0.45
0.1	$2 \leq T \leq 17$	0.47
0.3	$2 \leq T \leq 10$	0.49
0.5	$2 \leq T \leq 6$	0.53

Results III — the Central Charge from the Phase

5 · results



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$$\frac{\text{Im } \lambda_0}{T} = a_0 + \frac{B}{T} + \frac{C}{T^2} + \frac{D}{T^4} + \dots$$

dropped $\Rightarrow$ bias in c

Results IV — the Conformal Boundary Tower

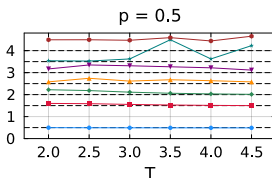
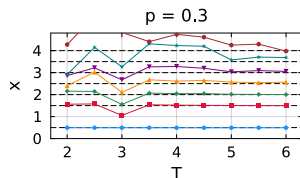
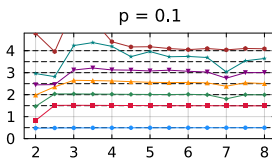
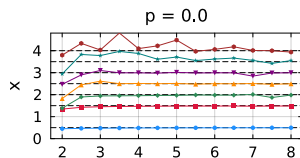
5 · results

Ising class, free boundaries: $x_\alpha \in \{\frac{1}{2}, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4\}$

identity tower + energy tower

$$\Delta\theta_\alpha = \arg \mu_\alpha - \arg \mu_0 = \frac{\pi V x_\alpha}{T}$$

block iteration with $k = 8$:
the eight leading eigenvalues



p	x_1
0	0.498
0.1	0.497
0.3	0.493
0.5	0.487

the conformal signatures in the Loschmidt echo
survive when integrability is broken

		from the eigenvectors		from the eigenvalues		
p	v_∞	window	c (entropy)	window	c (spectral)	x_1
0	2.00	$4 \leq T \leq 22$	0.49 ± 0.01	$2 \leq T \leq 22$	0.45	0.498
0.1	2.67	$4 \leq T \leq 13$	0.50 ± 0.04	$2 \leq T \leq 17$	0.47	0.497
0.3	3.97	$2 \leq T \leq 9$	0.53 ± 0.04	$2 \leq T \leq 10$	0.49	0.493
0.5	5.21	$2 \leq T \leq 5$	0.58 ± 0.04	$2 \leq T \leq 6$	0.53	0.487

c from **two dynamical routes** and **one equilibrium** + the **boundary content**

- ★ a translation-invariant 2nd-order MPO for the NNN chain
- ★ a block power method subleading spectrum of a non-Hermitian $\mathcal{E}$
- ★ the equilibrium inputs, measured v_∞ by exact diagonalisation, c by DMRG
- ★ the numerical simulations four couplings, local + cluster
- ★ the fits, windows and error analysis criteria for convergence of iteration methods

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open

why do the available T windows
close as p grows?

next

improve / optimise the block iteration
higher dimensions,
other universality classes and models

Thank you for your attention

:)



Backup A1 — from the Euclidean Strip to Real Time

backup · CFT

boundary CFT on a Euclidean strip of width ℓ , cut anchored at the edge:

$$S_2 = s_0 + \frac{c}{8} \log \left[\frac{2\ell}{\pi} \sin \frac{\pi x}{\ell} \right]$$

$c/8 = (c/12)(1 + 1/n)$ at $n = 2$ — halved, because an edge-anchored cut has one border and not two

continue to real time: $\ell \rightarrow i v T, \quad x \rightarrow i v t$

the ratio $x/\ell = t/T$ stays real $\Rightarrow$ the sine is untouched;
only the prefactor picks up $\log i = i\pi/2$

$$\operatorname{Re} S_2 = s_0 + \frac{c}{8} W(t, T) \quad (\text{dome}) \qquad \operatorname{Im} S_2 = \frac{c}{8} \cdot \frac{\pi}{2} = \frac{\pi c}{16} \quad (\text{plateau})$$

Backup A2 — the Spectrum, Derived I: from $T(z)$ to μ_i

backup · CFT

T : the **stress tensor** — the operator that generates conformal maps; its modes are the Virasoro generators L_n .

It is not primary: a map adds the inhomogeneous Schwarzian term, $\propto c$.

$$T(w) = \left(\frac{dz}{dw}\right)^2 T(z) + \frac{c}{12}\{z; w\}, \quad \{z; w\}_{\log} = -\frac{1}{2}\left(\frac{\pi}{\beta}\right)^2$$
$$\Rightarrow \hat{H} = \frac{\pi}{\beta}\left(L_0 - \frac{c}{24}\right) \quad \Rightarrow \quad \mu_i = \exp\left[-\frac{\pi}{\beta}\left(x_i - \frac{c}{24}\right)\right]$$

$\hat{H}$ is not T itself — it is T *integrated across* the strip: the generator of translations *along* it on the lattice, two non-universal additions — both the same for every level:

$$E_i = \underbrace{f\beta}_{\text{bulk}} + \underbrace{f_s}_{\text{boundary}} + \frac{\pi}{\beta}\left(x_i - \frac{c}{24}\right), \quad \log \mu_i = -E_i$$

level-independent terms cancel in every difference — differences isolate $x_i - x_0$

Backup A3 — the Spectrum, Derived II: the Expansion We Fit

backup · CFT

$$\beta = v(2\beta_0 + iT) = i v T \left(1 - \frac{2i\beta_0}{T}\right) \implies \frac{1}{\beta} = \frac{1}{i v T} \frac{1}{1 - \frac{2i\beta_0}{T}} = -\frac{i}{v T} \sum_{k \geq 0} \left(\frac{2i\beta_0}{T}\right)^k$$

a geometric series, and it converges:

$2\beta_0/T \leq 0.2$ in every run ($\beta_0 = 0.2$, $T \geq 2$)

k	goes to	order
0	phase	$1/T$
1	modulus	β_0/T^2
2	phase	β_0^2/T^3

$$\text{Im } \lambda_0 = A T + B + \frac{C}{T} + \mathcal{O}(T^{-3}), \quad A = -fv, \quad B = -\pi, \quad C = -\frac{\pi c}{24v}$$

no $1/T^2$ term (that order is real $\rightarrow$ modulus); B is pure branch (free fit: $B/\pi = -1.0000$)

the fit **truncates this expansion** $\Rightarrow$ part of the systematic on c

$$\frac{|\mu_i|}{|\mu_0|} = 1 - \frac{2\pi\beta_0(x_i - x_0)}{v T^2} \longrightarrow 1 \quad (\text{the circle})$$

Backup B1 — the Chain in Fermions: Where Integrability Breaks

backup · the model

$$H = - \sum_i \left[\sigma_i^z \sigma_{i+1}^z + \lambda \sigma_i^x + p (\sigma_i^z \sigma_{i+2}^z + \lambda \sigma_i^x \sigma_{i+1}^x) \right]$$

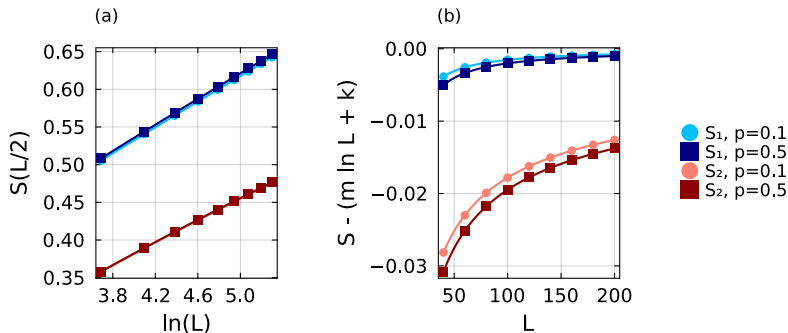
$$\text{Jordan-Wigner: } \sigma_j^x = 1 - 2n_j, \quad \sigma_j^z = (c_j + c_j^\dagger) K_j, \quad K_j = \prod_{k < j} (1 - 2n_k)$$

$$H = - \sum_i \left[\underbrace{(c_i^\dagger - c_i)(c_{i+1} + c_{i+1}^\dagger)}_{\text{NN: quadratic}} + \lambda \underbrace{(1 - 2n_i)}_{\text{field: quadratic}} + p \underbrace{(c_i^\dagger - c_i)(1 - 2n_{i+1})(c_{i+2} + c_{i+2}^\dagger)}_{\text{NNN: contains a quartic piece}} + p\lambda \underbrace{(1 - 2n_i - 2n_{i+1} + 4n_i n_{i+1})}_{\sigma^x \sigma^x: \text{density-density}} \right]$$

the p -terms carry four-fermion pieces $\Rightarrow$ Wick fails — no free-fermion solution for $p > 0$

$$S_n(L/2) = \frac{c}{12} \left(1 + \frac{1}{n}\right) \ln L + k_n + A_n L^{-x/n} \quad (\text{mid-chain cut, } x = 1 \text{ fixed})$$

$\Rightarrow c = 0.50$ at both couplings



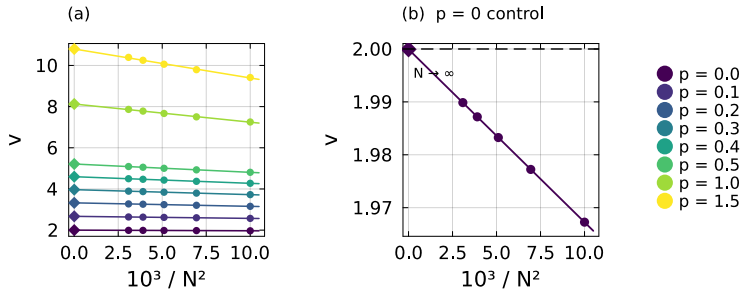
Backup B3 — Sound Velocity: Extrapolation

backup · the model

$$v(N) = \frac{N}{2\pi} [E_1(2\pi/N) - E_1(0)] \quad v(N) = v_\infty + \frac{C}{N^2}$$

empirical term: exponent fixed so the extrapolation reproduces $v = 2$ exactly at $p = 0$

the miss there: $2 \times 10^{-2} (1/N) \cdot 6 \times 10^{-5} (1/N^2) \cdot 6 \times 10^{-3} (1/N^3)$



Backup C1 — Schrödinger-Picture Benchmark of the MPO

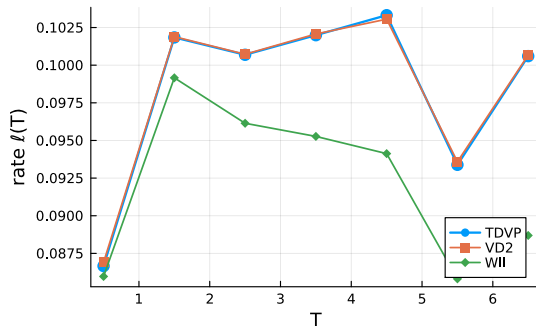
backup · the MPO

the Loschmidt rate we compute:

$$\ell(T) = - \lim_{N \rightarrow \infty} \frac{1}{N} \log |\mathcal{L}(T)|$$

finite chain, $N = 40$, $p = 0.1$:
repeated application of $U(\delta t)$ vs a TDVP
reference

W^I , W^{II} : the same machine, bond 4 (VD2: 13)
 W^I : disjoint products only
 W^{II} : + overlaps on one site
neither: two interactions across the same bond
 $\Rightarrow$ **first order** in δt



Zaletel et al. (2015) · Paeckel et al. (2019)

Backup C2 — the Overlap Only VD2 Can Host

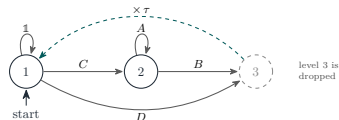
backup · the MPO

$$e^{\tau H} = \mathbb{1} + \tau \sum_x H_x + \frac{\tau^2}{2} \sum_{x,y} H_x H_y + \dots$$

the overlapping pairs are the τ^2 terms at stake

the concrete case: $\sigma_3^z \sigma_4^z$ and $\sigma_4^z \sigma_5^z$ share site 4

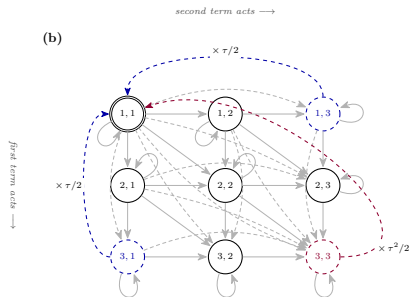
(a)



site 4 would need *B* and *C* — no such move exists:

the overlap is absent at τ^2

(b)



$$(1, 1) \xrightarrow{C} (2, 1) \xrightarrow{BC} (3, 2) \xrightarrow{B} (3, 3) \xrightarrow{\text{fold}} \frac{\tau^2}{2}$$

(swap the copies = the other ordering)

Backup C3 — Norm Drift Under Repeated Application

backup · the MPO

$|X+\rangle^{\otimes 100}$, $p = 0.1$, $\delta t = 0.05$, cutoff 10^{-12} after every layer

	W^I	W^{II}	VD2
virtual bond	4	4	13
$\ \psi\ $, 1 layer	1.3514	1.1341	0.99995
$\ \psi\ $, 8 layers	12.6146	3.0181	0.99923
$\ \psi\ $, 16 layers	307.3900	10.8328	0.99751

exact evolution is unitary, so any drift from 1 is pure construction error;
reaching $T = 10$ at this step takes ≈ 200 layers

Backup C4 — Convergence Orders, Measured

backup · the MPO

distance to the exactly diagonalised state ($N = 12$, $T = 1$, $p = 0.1$)

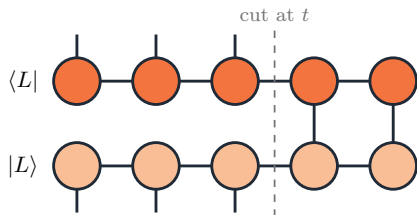
δt	W^I		W^{II}		VD2	
	error	order	error	order	error	order
0.2	2.01×10^1	—	2.54	—	1.09×10^{-1}	—
0.1	4.43	2.18	9.29×10^{-1}	1.45	2.44×10^{-2}	2.16
0.05	1.36	1.71	3.93×10^{-1}	1.24	5.93×10^{-3}	2.04
0.025	5.31×10^{-1}	1.35	1.80×10^{-1}	1.12	1.47×10^{-3}	2.01
0.0125	2.36×10^{-1}	1.17	8.65×10^{-2}	1.06	3.68×10^{-4}	2.00
0.00625	1.12×10^{-1}	1.08	4.24×10^{-2}	1.03	9.24×10^{-5}	2.00

$$\underbrace{\delta t^{q+1}}_{\text{one layer}} \times \underbrace{\frac{T}{\delta t}}_{\text{layers}} = T \delta t^q \implies \frac{\text{err}(2\delta t)}{\text{err}(\delta t)} = 2^q, \quad \log_2 \left[\frac{\text{err}(2\delta t)}{\text{err}(\delta t)} \right] = q$$

every layer applied in the Schrödinger picture — the operator's own construction error, isolated from the transverse contraction

Backup D1 — RDM vs RTM at a Temporal Cut

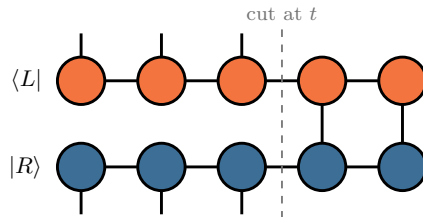
backup · truncation



$$(a) \rho_L = \text{Tr}_{T-t} |L\rangle \langle L|$$

pairs a state with **its own conjugate**

$\Rightarrow$ Hermitian, positive



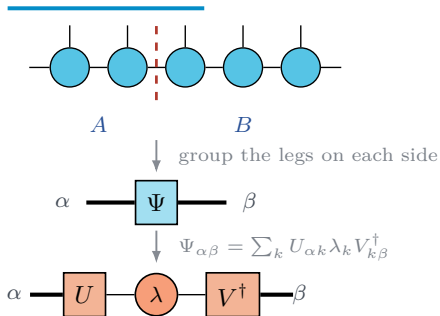
$$(b) \mathcal{T}_t = \text{Tr}_{T-t} |R\rangle \langle L| / \langle L|R\rangle$$

pairs the **two distinct fixed points** $\langle L|, |R\rangle$

$\Rightarrow$ non-Hermitian, complex eigenvalues
allowed

Backup D2 — Entanglement through the SVD

backup · truncation



$$|\Psi\rangle = \sum_{ij} \Psi_{ij} |i\rangle_A |j\rangle_B$$

across the cut the state is a matrix

$$\Psi = U\lambda V^\dagger \implies |\Psi\rangle = \sum_k \lambda_k |v_k\rangle_A |w_k\rangle_B$$

the **Schmidt decomposition**

$$S = -\text{Tr}(\rho_A \log \rho_A) = -\sum_k \lambda_k^2 \log \lambda_k^2 \quad \text{the } \lambda_k^2 \text{ are the eigenvalues of } \rho_A$$

one set of numbers ranks the directions *and* carries the entanglement

$\Rightarrow$ truncation = keep the largest λ_k

Backup D3 — the Local RDM Reduction

backup · truncation

$$\begin{array}{c}
 \begin{array}{c} A \\ A^* \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} = \left(\begin{array}{c} \text{blue} \\ \text{blue} \end{array} \right) \quad (a)
 \end{array}$$

$$\begin{array}{c}
 \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{orange} \\ \text{orange} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \\
 M \text{ traced}
 \end{array}
 = EIG^{1/2} \left(\begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{orange} \\ \text{orange} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \right) = EIG^{1/2} \left(\begin{array}{c} U \\ U^\dagger \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{orange} \\ \text{orange} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \right) = SV \left(\begin{array}{c} \text{orange} \\ \text{orange} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \right) = SV \left(\begin{array}{c} \text{orange} \\ \text{orange} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \begin{array}{c} \text{blue} \\ \text{blue} \end{array} \right) \quad (b)$$

- ▶ mixed canonical form: away from the cut every tensor meets **its own conjugate** and contracts to the identity — panel (a)
- ▶ the traced block then disappears column by column and the open block groups into one isometry per side: $\rho_B = U_B M_A U_B^\dagger$ with $U_B^\dagger U_B = \mathbb{1} \Rightarrow \text{eig}(\rho_B) = \text{eig}(M_A)$, a $\chi \times \chi$ object — the exponentially large matrix is never built
- ▶ eigenvalues = squared singular values of M , one-to-one $\Rightarrow$ keep the χ' largest (optimal; error = discarded weight)

Backup D4 — RTM Truncation: the Same Reduction, Different Isometries

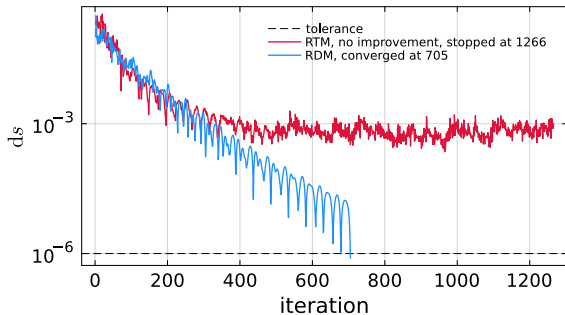
backup · truncation

- ▶ generalised canonical form: each $\langle L|$ tensor is paired with the corresponding $|R\rangle$ tensor — the isometries on the two sides differ
- ▶ local reduction: $\tau_B = U_B M_A V_B$ with $U_B \neq V_B^\dagger \Rightarrow$ eigenvalues NOT preserved, but $\text{sv}(\tau_B) = \text{sv}(M_A)$
- ▶ so we truncate the $\chi \times \chi$ environment on its **singular values**; for the RTM these are no longer one-to-one with the eigenvalues (non-Hermitian) — no variational bound

Backup D5 — How the Iteration Stops

backup · truncation

1. ds = change in the singular values of the temporal state between successive iterations — the convergence criterion in the code
2. stop when ds falls below tolerance, or after 150 iterations without improvement
3. RTM hits a noise floor (no-improvement stop) while RDM converges — yet both return the same leading eigenvalue to better than 1 part in 10^4



the two routes need different things from $\mathcal{E}$:

eigenvalues: second-order accurate

eigenvectors: sensitivity $\propto 1/\text{gap}$

emergent dual unitarity shrinks the gaps $\Rightarrow$ the eigenvectors go first
first rejected run at $T = 18, 11, 7, 3.5$ as p goes $0 \rightarrow 0.5$

is that just a *closing* gap? — exact diagonalisation, no truncation, no tensor networks:

p	T	dimension	$ \mu_0 - \mu_1 $	$\text{cond}(V)$
0	3.5	2187	0.21	3.0×10^2
0.1	1.5	2197	0.30	2.2×10^{11}
0.3	1.5	2197	0.19	2.1×10^{10}
0.5	1.5	2197	0.14	9.6×10^9

the gap barely moves; the eigenvector *basis* degenerates by 7–9 orders

Algorithm (App. F, Alg. 1): the k dominant eigenpairs of $\mathcal{E}$

1. $|R_j\rangle, \langle L_j| \leftarrow$ random complex tMPS, $j = 1, \dots, k$
2. **apply:** $|\tilde{R}_j\rangle \leftarrow \text{truncate}_\chi(\mathcal{E} |R_j\rangle), \quad \langle \tilde{L}_j| \leftarrow \text{truncate}_\chi(\langle L_j| \mathcal{E})$
3. **project:** $S_{ij} = \langle L_i | R_j \rangle, \quad M_{ij} = \langle L_i | \mathcal{E} | R_j \rangle$
4. solve $Mv = \theta Sv$ (via $\text{pinv}(S)M$) for Ritz values $\theta_1, \dots, \theta_k$ and coefficients V, U
5. **separate:** $|R_j\rangle \leftarrow \text{truncate}_\chi(\sum_i V_{ij} |\tilde{R}_i\rangle)$, analogously $\langle L_j|$ with U
6. normalise; repeat 2–5 until the leading Ritz values stop changing within tolerance

Backup E2 — the Block Method against Exact Diagonalisation

backup · the block method

at short times the rotated column fits as a dense array: diagonalise it with no state truncation and no projection onto a block, and compare

operator	dim.	exact $ \mu_0 , \mu_1 $	error	stopped on
critical Ising, $T = 1.5$	27	0.9048, 0.8373	2.5×10^{-12}	tolerance
critical Ising, $T = 2$	81	0.9129, 0.8702	2.3×10^{-7}	no improvement
ANNNI-type $p = 0.1$, $T = 1.5$	2197	0.9405, 0.9082	1.2×10^{-7}	no improvement

$\delta t = 0.5$, no cooling sites; none of the three reaches the bond cap

**the reduced eigenproblem adds no measurable error —
the accuracy is set by *convergence*, not by the projection**

Backup F1 — Why the Spectral c Sits Below $\frac{1}{2}$ at $p = 0$

backup · the fits

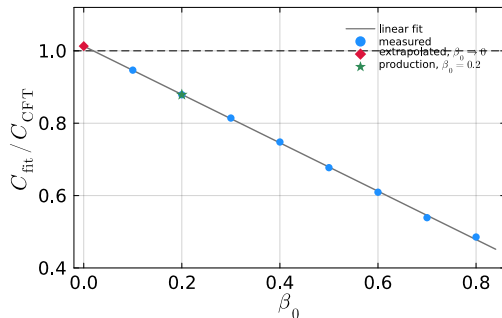
1. the fit truncates the expansion

$$\frac{\text{Im } \lambda_0}{T} = a_0 - \frac{\pi}{T} - \underbrace{\frac{\pi c}{24 \nu T^2}}_{c \text{ from here}} + \underbrace{\frac{D}{T^4}}_{\text{dropped}} + \dots$$

dropped term = D/T^3 in $\text{Im } \lambda_0$

restoring it: 90% $\rightarrow$ 97% of the predicted coefficient

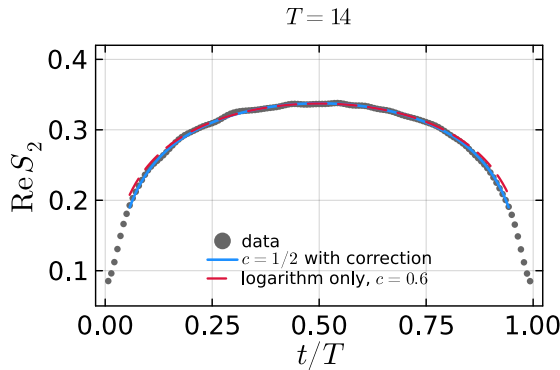
2. the regulator β_0 is finite



Backup F2 — Why the Real Part Cannot Measure c

backup · the fits

$$\operatorname{Re} S_2 = s_0 + \frac{c}{8} W + a w^{-1/2}$$



Backup F3 — Four Boundary Condition Pairs

backup · the fits

a **tower** = the set of boundary operators a given pair of boundary conditions allows, and their descendants

